

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456715.72851 +/- 0.0007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.017
Transit depth (Rp/Rs)^2	2.96 +/- 0.59 [%]
Orbital Inclination (inc)	80.84 +/- 1.92
Airmass coefficient 1 (a1)	1.181 +/- 0.01
Airmass coefficient 2 (a2)	-0.1146 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	3.86
Optimal Annulus	8.86
Transit Duration (day)	0.0456 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.172 ± 0.017 vs 0.1594 (deviation 0.74σ)
Duration fit vs published	0.0456 d vs 0.0512 d (deviation 0.64σ)
Mid-time O–C	1.6 min
Depth SNR	10.1
Residual scatter	0.87 %

Light curve

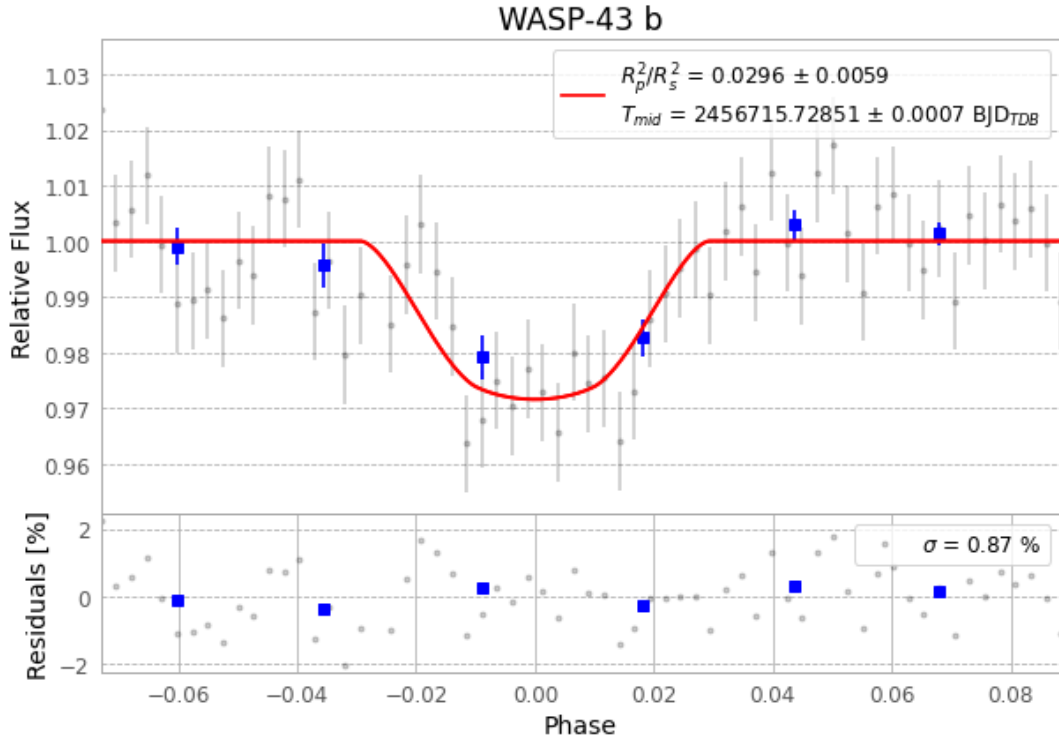


Figure 1 — FinalLightCurve_WASP-43 b_2014-02-26.png (SHA-256 c44070b0113cb647...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [141, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 06de6e25cbf1e1bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140227035412.FITS → WASP-43140227070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140227.log (0e8ebde85c6d...), FinalLightCurve_WASP-43 b_2014-02-26.png (c44070b0113c...), FinalLightCurve_WASP-43 b_2014-02-26.csv (3f8054b099c9...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 01:10Z.